



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°5_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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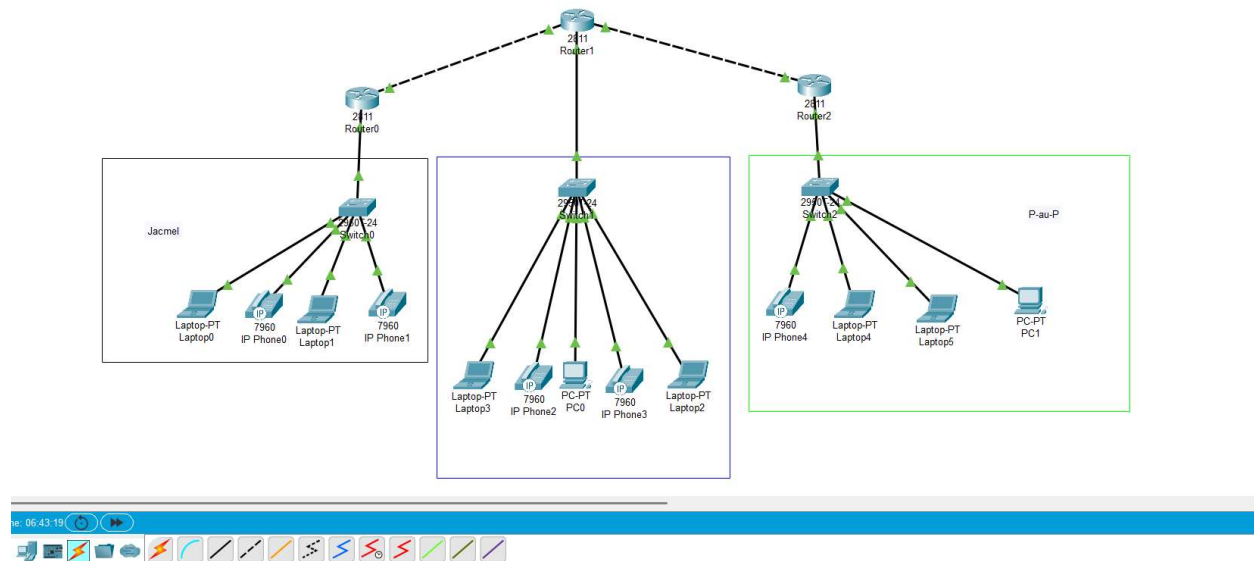
NIVEAU: L3

DATE: 27/04/2026

L'objectif de ce TD est de:

1. Déployer la VoIP avec Call Manager Express
2. Assurer la communication entre sites
3. Mettre en place DHCP + routage
4. Tester appels inter-sites

Mettons en place un réseau VoIP multi-sites (Jacmel – Cayes – Port-au-Prince)



Configuration R1

Configuration de base

Press RETURN to get started!

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#interface FastEthernet0/1
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
```

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☐ Top

DHCP VoIP

```
Router(config-if)#ip dhcp pool VLAN20
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# option 150 ip 192.168.20.1
Router(dhcp-config)#
```

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CME

```
Router(dhcp-config)#telephony-service
Router(config-telephony)# max-ephones 5
Router(config-telephony)# max-dn 5
Router(config-telephony)# ip source-address 192.168.20.1 port
2000
Router(config-telephony)# auto assign 1 to 5
Router(config-telephony)#exit
Router(config)#
```

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] Top

Extensions

```
Router(config)#ephone-dn 1
Router(config-ephone-dn)# number 2001
Router(config-ephone-dn)#
Router(config-ephone-dn)#ephone-dn 2
Router(config-ephone-dn)# number 2002%LINK-3-UPDOWN: Interface
ephone_dsp DN 1.1, changed state to up
%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up

Router(config-ephone-dn)#
```

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Configuration R2

Configuration de base

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)# ip address 192.168.30.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# ^
% Invalid input detected at '^' marker.

Router(config-if)# ip address 10.0.0.6 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#interface FastEthernet1/0
Router(config-if)# ip address 10.0.0.10 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

```

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DHCP VoIP

```

Router(config-if)#ip dhcp excluded-address 192.168.30.1
192.168.30.10
Router(config)#ip dhcp pool VLAN30
Router(dhcp-config)# network 192.168.30.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.30.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# option 150 ip 192.168.30.1
Router(dhcp-config)#

```

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CME

```
Router(dhcp-config)#telephony-service
Router(config-telephony)# max-ephones 5
Router(config-telephony)# max-dn 5
Router(config-telephony)# ip source-address 192.168.30.1 port
2000
Router(config-telephony)# auto assign 1 to 5
Router(config-telephony)#exit
Router(config)#
```

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Extensions

```
Router(config)#ephone-dn 1
Router(config-ephone-dn)# number 3001
Router(config-ephone-dn)#
Router(config-ephone-dn)#ephone-dn 2
Router(config-ephone-dn)# number 3002%LINK-3-UPDOWN: Interface
ephone_dsp DN 1.1, changed state to up
%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up
Router(config-ephone-dn)#
```

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Configuration R2

Configuration de base


```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)# ip address 192.168.40.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#interface FastEthernet0/1
Router(config-if)# ip address 10.0.0.14 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#
```

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DHCP VoIP

```
Router(config-ephone-dn)#ip dhcp excluded-address 192.168.40.1
192.168.40.10
Router(config)#ip dhcp pool VLAN40
Router(dhcp-config)# network 192.168.40.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.40.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# option 150 ip 192.168.40.1
Router(dhcp-config)#
```

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CME

```
Router(dhcp-config)#telephony-service
Router(config-telephony)# max-ephones 2
Please remove ephone 3 to 5 by re-configuring or reloading the
system!!
Router(config-telephony)# max-dn 2
Router(config-telephony)# ip source-address 192.168.40.1 port
2000
Router(config-telephony)# auto assign 1 to 2
Router(config-telephony)#
```

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Extensions

```
Router(config-telephony)#ephone-dn 1
Router(config-ephone-dn)# number 4001
```

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Configuration des commutateurs VLAN VoIP

S1


```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5,  
changed state to up
```

```
Switch>enable  
Switch#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#hostname S1  
S1(config)#interface range fa0/1-23  
S1(config-if-range)# switchport mode access  
S1(config-if-range)# switchport access vlan 10  
% Access VLAN does not exist. Creating vlan 10  
S1(config-if-range)# switchport voice vlan 10  
S1(config-if-range)#exit  
S1(config)#
```

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S2

```
changed state to up
```

```
Switch>enable  
Switch#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#hostname S2  
S2(config)#interface range fa0/1-23  
S2(config-if-range)# switchport mode access  
S2(config-if-range)# switchport access vlan 20  
% Access VLAN does not exist. Creating vlan 20  
S2(config-if-range)# switchport voice vlan 20  
S2(config-if-range)#
```

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S3

changed state to up

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S3
S3(config)#interface range fa0/1-23
S3(config-if-range)# switchport mode access
S3(config-if-range)# switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
S3(config-if-range)# switchport voice vlan 30
S3(config-if-range)#exit
S3(config)#
```

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Routage entre les trois routeurs (OSPF)

R1

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.4 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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R2

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.8 0.0.0.3 area 0
Router(config-router)# network 10.0.0.12 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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R3

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.16 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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Dial-peer (permet appels entre sites)

Sur R1

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#dial-peer voice 1 voip
Router(config-dial-peer)# destination-pattern 1...
Router(config-dial-peer)# session target ipv4:192.168.10.1
Router(config-dial-peer)#dial-peer voice 2 voip
Router(config-dial-peer)# destination-pattern 3...
Router(config-dial-peer)# session target ipv4:192.168.30.1
Router(config-dial-peer)#
```

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Sur R2

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#dial-peer voice 1 voip
Router(config-dial-peer)# destination-pattern 1...
Router(config-dial-peer)# session target ipv4:192.168.10.1
Router(config-dial-peer)#dial-peer voice 2 voip
Router(config-dial-peer)# destination-pattern 2...
Router(config-dial-peer)# session target ipv4:192.168.20.1
Router(config-dial-peer)#dial-peer voice 3 voip
Router(config-dial-peer)# destination-pattern 3...
Router(config-dial-peer)# session target ipv4:192.168.30.1
Router(config-dial-peer)#
```

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Sur R3

```
Router(config-dial-peer)#dial-peer voice 1 voip
Router(config-dial-peer)# destination-pattern 1...
Router(config-dial-peer)# session target ipv4:192.168.10.1
Router(config-dial-peer)#dial-peer voice 2 voip
Router(config-dial-peer)# destination-pattern 2...
Router(config-dial-peer)# session target ipv4:192.168.20.1
Router(config-dial-peer)#
Router(config-dial-peer)#dial-peer voice 3 voip
Router(config-dial-peer)# destination-pattern 3...
Router(config-dial-peer)# session target ipv4:192.168.30.1
```

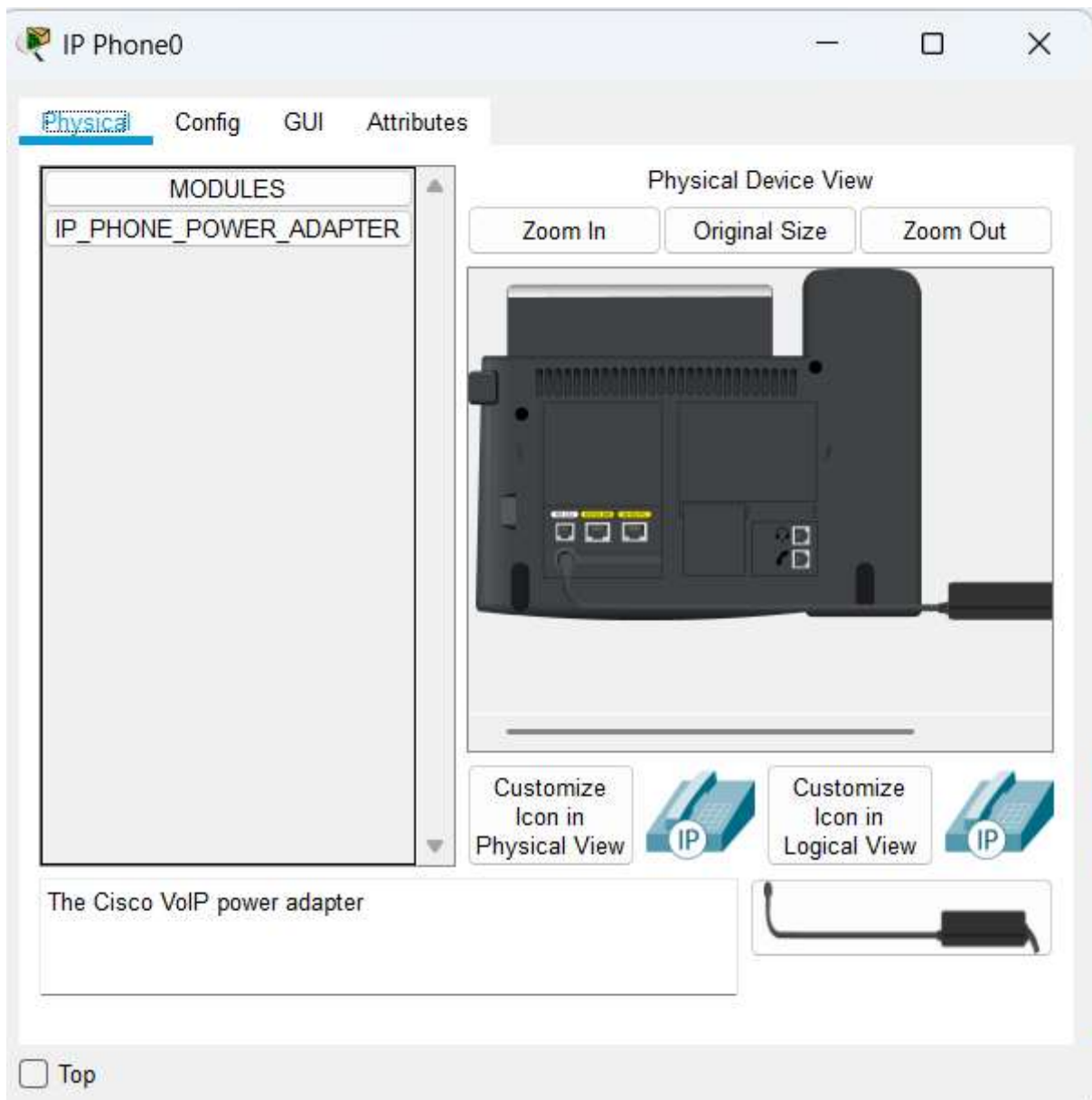
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Activation d'un telephone



Test ping


```
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time=2ms TTL=125
Reply from 192.168.30.2: bytes=32 time=2ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>
```

Questions

1. Quelle est la différence entre la VoIP et la téléphonie classique ?

La VoIP se distingue de la téléphonie classique car elle convertit la voix en paquets IP transmis sur un réseau Internet, tandis que la téléphonie traditionnelle utilise des circuits commutés dédiés pour chaque appel.

2. Quel est le rôle du dial-peer dans un réseau VoIP ?

Le dial-peer joue un rôle essentiel dans un réseau VoIP puisqu'il définit la manière dont un appel est acheminé, en précisant la source, la destination et les paramètres techniques comme le codec ou la qualité de service.

3. Pourquoi utilise-t-on OSPF dans un réseau VoIP multi-site ?

Dans un réseau VoIP multi-site, OSPF est utilisé pour assurer un routage rapide et fiable entre les différents sites, ce qui garantit la continuité et la qualité des communications même en cas de panne ou de changement de chemin.

Coclusion.

Ce TD m'a permis de savoir comment deployer la VoIP avec Call Manager Express, Mettre en place DHCP + routage et tester les appels inter-sites .donc la tache a été réussie.